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# Vacuum ultraviolet measurements on hydrogen resonance lines in the Maryland Centrifugal Experiment

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Extended spectroscopic measurements of hydrogen emission into the vacuum ultraviolet region on the Maryland Centrifugal Experiment [R. F. Ellis, A. B. Hassam, S. Messer, and B. R. Osborn, *Phys. Plasmas* **8**, 2057 (2001)] indicate, from opacity on the Lyman series,  $\sim 6\%$  atomic hydrogen coexisting with protons, electrons, and carbon-impurity ions in a plasma region where the temperature is  $\sim 15$  eV. An explanation is offered, based upon charge exchange of entering cold hydrogen atoms with plasma protons, followed by collisional excitation and radiative decay. © 2005 American Institute of Physics. [DOI: 10.1063/1.1854687]

In a previous paper,<sup>1</sup> visible spectroscopy was used as a diagnostic on the Maryland Centrifugal Experiment (MCX). In this device,<sup>2</sup> enhanced confinement along axial magnetic field lines in a solenoid with mirror end fields is sought by applying a radial electric field between a central conductor and the outer wall, thereby generating azimuthal  $\vec{E}_r \times \vec{B}_z$  forces. This configuration offers reduced end losses, shear stabilization,<sup>3</sup> and viscous heating.<sup>4</sup> Observations along a chord in the plasma of the Doppler shift of spectral lines yielded a confirmation of rotation of the plasma, with azimuthal velocities ranging from 20 to 100 km/s for carbon impurities in increasing ionization states and 10 km/s for hydrogen atoms. Also obtained from thermal Doppler line broadening were impurity-ion temperatures of 10–40 eV, and hydrogen-atom temperatures of 5–10 eV. An electron temperature of  $\sim 15$  eV was inferred from relative line intensities of different ionic species of carbon.

In an effort towards data refinement, and a further understanding of these differences (possibly related to spatial variations), we expanded the hydrogen measurements from the  $n \rightarrow 2$  Balmer series in the visible to the  $n \rightarrow 1$  Lyman series in the vacuum-ultraviolet (vuv) spectral region. The widths of the lower- $n$  Lyman resonance lines are expected to be enhanced by opacity, decreasing towards higher quantum number along the series, with an associated increase in the depth of view. Such opacity-sensitive measurements can be very useful for comparison with radiative transfer modeling in fusion devices and, as herein, for determination of the absorber lower (ground) state density.

In the MCX, the diameter of the vacuum vessel is 55 cm at the midplane, with separations of 250 cm between the two mirror coils and 330 cm between the two insulators which intersect the magnetic field lines outside the mirror planes. The axial core consists of a hollow stainless steel tube with an outer diameter of 4.8 cm. Typical operation for these experiments was with a midplane magnetic field of 0.2 T (2 kG) and a mirror magnetic field of 1.8 T (18 kG), for a mirror ratio of 9. The initial potential between the core and the wall was  $-7$  kV. The hydrogen gas fill was at a pressure

of 5 mTorr ( $3 \times 10^{14}$  atoms/cm<sup>3</sup>). A typical shot lasted  $\sim 7$  ms. The plasma was found to exist in two distinct modes. The first “high-rotation” mode lasted  $\sim 2.5$  to 3 ms (depending upon the magnetic field, the radial electric field, and the external resistance of the circuit) and was characterized by low radial currents, high rotational velocities, and good confinement times, but also by an abundance of impurity lines. The presence of such impurities presumably resulted from a release at the wall, at the central conductor, and at the end insulators. The spontaneous onset of the second “ordinary-rotation” mode at  $\sim 3$  ms after breakdown was accompanied by a greatly reduced impurity level and improved signal-to-noise ratio for the higher- $n$  Lyman-series spectral lines. Hence, for most of the measurements reported here, the detector was gated on for a time of 3–7 ms.

For the vacuum ultraviolet spectroscopic measurements, the plasma was viewed along a chord 16.5 cm from the axis of symmetry, as indicated in Fig. 1. A 1 m,  $f/10$ , normal incidence monochromator equipped with a 1200 grooves/mm grating blazed at  $2.7^\circ$  (80 nm Littrow wavelength) was used in first order. It was custom coated with magnesium fluoride for maximum efficiency in the 93.0–121.6 nm wavelength range of interest. The reciprocal linear dispersion of the spectrograph was 0.83 nm/mm at the focal plane, where an evaporated layer of Liumogen<sup>5</sup> phosphor served to convert the vacuum ultraviolet radiation into broadband fluorescent emission at a wavelength centered at 530 nm. The image at this phosphor was focused with a magnification of 2.8 times onto a microchannel-plate-intensified linear diode array with a pixel size of 25  $\mu\text{m}$ . The resulting dispersion at the detector was 135 pixel/nm, as measured directly. With a total of 900 pixels, the overall wavelength coverage was 6.7 nm.

An instrumental resolution of 6 pixels or 0.044 nm was determined by combining numerical best fittings of two Gaussian profiles to two C II impurity lines at 103.634 and 103.702 nm, i.e., separated by 0.068 nm. This resolution is consistent with the width of a reabsorption dip appearing (particularly when viewing in the early phase) near the center

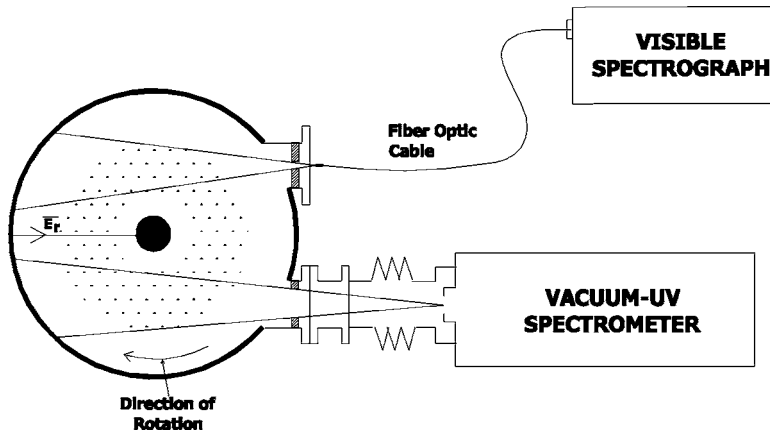


FIG. 1. A cross section schematic (not to scale) of the MCX vacuum chamber with a central electrode. Both the vacuum ultraviolet and visible spectrographs are shown in an unfocused mode, with an  $\sim 14$  cm radial view at midplane. Implied distinct radial limits for the plasma are symbolic, representing a  $\sim 19$  cm wide plasma view.

of the Lyman- $\alpha$  line of neutral hydrogen at a wavelength of 121.6 nm, as discussed below.

The relative sensitivity of the vacuum ultraviolet spectrograph was determined for the  $n \rightarrow 1$  Lyman-series lines at wavelengths between 93.1 nm ( $n=7$ ) and 102.5 nm ( $n=3$ ) by comparison with the  $n \rightarrow 2$  Balmer-series lines at wavelengths between 397.0 nm ( $n=7$ ) and 656.3 nm ( $n=3$ ), which were calibrated using a tungsten lamp as described below. Here the branching ratios of transition probabilities<sup>6</sup> were used for the comparison of those vuv lines least affected by opacity (see Table I). This relative calibration was continued over the series by comparison with a capillary discharge lamp, which was also cross calibrated with the Balmer series. For the most intense Lyman- $\alpha$  line, a ND=0.8 neutral-density filter of 16% transmission was added in the fluorescent beam. In order to avoid variations of sensitivity across the phosphor and the detector, each spectral line was recorded at the same location on the phosphor and the detector surfaces by varying the grating angle between shots.

With no focusing lens at the entrance of the vuv spectrograph, the lateral extent of the plasma viewed at the midplane was 14 cm. The view was aperture limited by the port-hole through the vacuum tank (see Fig. 1). As such, the view was of the more dense portion of the  $\sim 19$  cm radial extent of the plasma. As illustrated in Fig. 1, this wide view was also used with the visible spectrometry described next. An alternate focused view in the visible was also used for a comparison with the previous measurements.<sup>1</sup> With a possible excep-

tion at the center of the optically-thick vuv lines, the emissions measured were an average along the chordal line of sight.

Supporting visible spectroscopy evolved from previous measurements<sup>1</sup> of the Balmer-series lines, with the addition of a measurement of electron density from Stark broadening. In the present case, observations were possible for lines with wavelengths from 656.3 nm for  $n=3$  to 379.8 nm for  $n=10$ . The lines were dispersed using a 1.0 m Czerny–Turner stigmatic spectrograph. The plasma was viewed simultaneously through a quartz fiber along a similar axis as that for the vacuum-uv spectroscopy, except at the upper port (see Fig. 1). Initially a 1 cm diameter focusing lens was used to reproduce the earlier<sup>1</sup> data with a viewing aperture limited to  $\lesssim 5$  mm in the plasma. However, in an effort to better compare the visible and vuv data, the visible spectra were recorded here without a focusing lens (as shown in Fig. 1) in order to provide essentially the same wide field of view as that of the vuv spectrometer.

A charge coupled device camera, intensified and gated by a lens-coupled microchannel plate, was used to record the visible emission. The size of each detector element (pixel) was 25  $\mu\text{m}$ , with 748 active elements in the wavelength direction. A 1200 grooves/mm grating provided a reciprocal dispersion near 600 nm in first order of 0.0178 nm/pixel. This resulted in a 11.8 nm coverage of the spectrum. The instrumental linewidth was limited by the detector resolution and was measured to be 3 pixels (75  $\mu\text{m}$ ) using a 50  $\mu\text{m}$

TABLE I. Measured and calculated parameters and vuv linewidths (in nm).

| Lyman      | $n$ | $\lambda(\text{nm})$ | $w_m^a$ | $w_{\text{ins}}$ | $w_D$ | $w_{\text{Stk}}$ | Reference     | $\tau^b$ |
|------------|-----|----------------------|---------|------------------|-------|------------------|---------------|----------|
| $\alpha$   | 2   | 121.57               | 0.082   | 0.044            | 0.036 | 0.00090          | Ref. 7        | 20       |
| $\beta$    | 3   | 102.57               | 0.089   | 0.044            | 0.031 | 0.0044           | Refs. 7–9     | 3.1      |
| $\gamma$   | 4   | 97.25                | 0.112   | 0.044            | 0.029 | 0.0081           | Ref. 7        | 1.1      |
| $\delta$   | 5   | 94.97                | 0.122   | 0.044            | 0.028 | 0.013            | Refs. 8 and 9 | 0.5      |
| $\epsilon$ | 6   | 93.78                | 0.111   | 0.044            | 0.028 | 0.019            | $\propto n^2$ | 0.29     |
| $\zeta$    | 7   | 93.07                | 0.090   | 0.044            | 0.028 | 0.025            | $\propto n^2$ | 0.18     |
| $\iota$    | 8   | 92.62                | 0.090   | 0.044            | 0.028 | 0.033            | $\propto n^2$ | 0.12     |

<sup>a</sup>Measured widths  $\pm 15\%$ – $20\%$ .

<sup>b</sup>For  $d=30$  cm,  $T_H=15$  eV,  $N_H=4.6 \times 10^{13} \text{ cm}^{-3}=(0.06)N_e$ .

entrance slit width, for a spectral resolution of 0.053 nm. The effect of the instrumental line broadening was included by convoluting a representative Lorentzian profile with a simulated Gaussian profile for the Doppler-broadened plasma emission and achieving a best fit. The relative sensitivity of the spectrograph was determined by calibration with a tungsten ribbon standard lamp, as mentioned above. This was possible over a wavelength range from 656.3 nm at Balmer  $\alpha$  ( $n=3$ ) to 397.0 nm at Balmer  $\varepsilon$  ( $n=7$ ), as limited by the output lamp. This provided relative Balmer line intensities useful in our interpretations which follow.

The measured  $n \rightarrow 1$  Lyman series full widths at half maximum for  $n=2-8$  (Lyman  $\alpha$  to  $\zeta$ ) are listed as  $w_m$  in column 4 of Table I, with an estimated accuracy varying from  $\pm 15\%$  for Lyman  $\alpha$  to  $\pm 20\%$  for Lyman  $\zeta$ , and are found to be approximately all equal. In all cases they exceed the value for the instrumental width of 0.044 nm given above and included in column 5 of Table I for comparison. Such similar widths, when the Doppler broadening is decreasing for the high- $n$  lines (column 6), can be understood by an increase in Stark broadening along the series (column 7 and references shown). Also, opacity broadening is more important on the lower- $n$  lines, which have greater line strengths<sup>6</sup> and therefore opacity  $\tau$ , as shown in column 8. While the most intense Lyman- $\alpha$  line was relatively free from underlying impurity contributions, the increasingly less intense recorded spectra for the higher- $n$  lines included significant narrowband noise, originating from impurity lines from carbon, oxygen, and metals such as copper and nickel, which affected somewhat the fitting to the observed widths.

In order to better understand the processes responsible for the observed Lyman line data, especially concerning opacity, a series of best-fit convolutions were performed for each line in the series, with the requirement that they be self-consistent within the series. Another requirement was that the results be consistent with those from the Balmer series. In fact, the widths of the higher members of the Balmer series were significantly larger than could be explained by a thermal Doppler effect convoluted with an instrument function, and were used to derive an electron density of  $N_e = (7.7 \pm 0.8) \times 10^{14} \text{ cm}^{-3}$ , based upon computations for both  $n=5$  (Ref. 7) and  $n=6$  (Refs. 8 and 9) and an experiment<sup>10</sup> for  $n=8$ . This density is in agreement with an interferometric measurement.<sup>11</sup> It is somewhat larger than the hydrogen-atom fill density of  $3 \times 10^{14} \text{ cm}^{-3}$ , which is thought to be associated with input from vapor near the wall.

The  $n=4$  to  $n=1$  Lyman- $\gamma$  line was selected as a beginning of the vuv analysis, because it was relatively unaffected by both Stark broadening and opacity (see Table I). A Gaussian profile of width 0.029 nm simulating the Doppler-broadened plasma emission was convoluted with a 0.044 nm wide Lorentzian instrument function to give the best fit to the measurements. Actually this Doppler width was adjusted also for a best fit to the profile of the Lyman- $\delta$  line, the last in the series for which Stark calculations<sup>8,9</sup> were available, and then on to the higher- $n$  lines by extrapolating the theoretical Stark widths as  $n^2$  (see Table I). The fact that these higher- $n$  Lyman lines can be fitted with Stark broadening, which is sensitive to electron density, supports our measure-

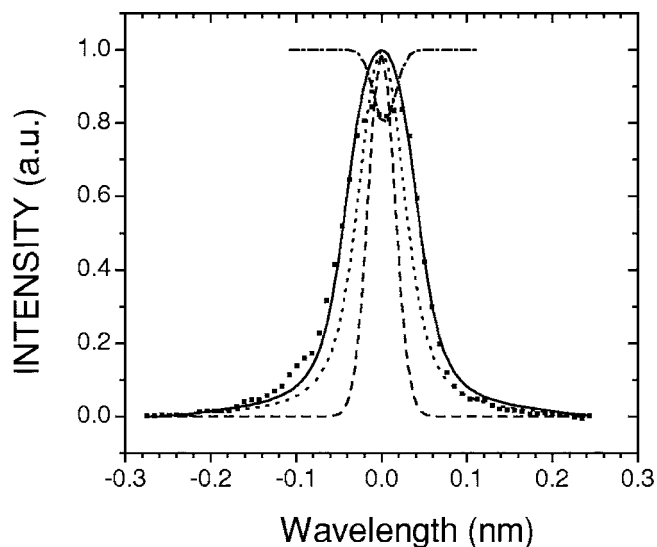


FIG. 2. Best fit for the Lyman- $\alpha$  line of a Gaussian profile (dashed) simulating Doppler broadening, also shown convoluted (dotted) with a 0.044 nm wide Lorentzian instrument function, and finally with opacity added to the initial Gaussian and again convoluted with the instrument function (solid curve). The data are shown as solid squares representing individual pixels. The absorption dip near line center is simulated by a (negative) Gaussian, plotted with alternating dots and dashes.

ment of that density from the higher Balmer lines, as described above. The best Doppler fit to the Lyman- $\gamma$  (and higher- $n$ ) lines corresponds to a hydrogen temperature of  $T_H \approx 15 \text{ eV}$ . This new and increased temperature value is consistent<sup>1</sup> both with a measured ion temperature of  $T_i \approx 10-12 \text{ eV}$  for  $C^+$  ions and an electron temperature of  $T_e \approx 15 \text{ eV}$ , and will be discussed further below.

For the  $n=2-1$  Lyman- $\alpha$  line, the Doppler width scales linearly upward with wavelength to 0.036 nm and clearly would not provide a suitable fit for this intense line, even when convoluted with the instrument function, as shown in Fig. 2. Additional opacity broadening was applied to modify the Doppler profile by using the basic radiative transfer relation  $I_{21}(\lambda) \propto 1 - e^{-\tau(\lambda)}$  for a uniform medium. Here  $\tau(\lambda)$  represents the opacity, i.e., the product of the line absorption coefficient and the depth of the medium  $d$ . As such,  $\tau(\lambda)$  varies linearly with the neutral hydrogen density  $N_H$  (to be determined), the oscillator strength<sup>6</sup>  $f=0.416$  and the wavelength of the line and inversely with the square root of the hydrogen-atom temperature, along with the Gaussian line shape function.<sup>12</sup> The resulting opacity-broadened Gaussian profile  $I_{21}(\lambda)$  was again convoluted with a Lorentzian instrument function. As shown in Fig. 2, it was found that a best overall fit to the data was obtained for a peak opacity of  $\tau(\lambda_0) \sim 20$  which leads, for an assumed depth  $d=30 \text{ cm}$  and a measured temperature  $T_H=15 \text{ eV}$ , to  $N_H=4.6 \times 10^{13} \text{ cm}^{-3}$ , which is 6% of the measured electron density. (The plasma may have a larger radial extent than suggested by the illustration in Fig. 1.)

On some shots, particularly during the early mode following breakdown, the Lyman- $\alpha$  line emitted from the rotating plasma showed a near-central dip, characteristic of absorption in a thin and dense, cold, stationary outer layer near the wall. This dip can be seen in Fig. 2 where a correction to



the peak intensity has been applied, based upon a Gaussian best fit to the absorption dip. Small redshifts of the center of the plasma line emission from that of the reabsorption dip were observed on some shots taken during the early phase of the discharge and were found to be consistent with earlier measurements<sup>1</sup> of a 5.4 times larger (scaling linearly with wavelength) Doppler shift of the Balmer- $\alpha$  line associated with plasma rotation, as indicated in Fig. 1.

For the  $n=3$  to  $n=1$  Lyman- $\beta$  line, the peak opacity assumed was reduced to 3.1, because of the lower oscillator strength and the shorter wavelength compared to that of the Lyman- $\alpha$  line. No definite central dip associated with any considerably-weaker absorption near the wall was observed. Nevertheless, there was a noticeable enhancement of the calculated width when opacity is included. Also apparent were the contributions of copper and nickel impurity lines, which became more apparent at the rather sharply reduced grating efficiency at wavelengths between the Lyman- $\alpha$  and Lyman- $\beta$  lines.<sup>5</sup>

The existence of hydrogen atoms radiating on Lyman transitions from a plasma at a temperature of  $T_H \approx 15$  eV, as encountered here, can occur when cold atoms enter the plasma and undergo symmetrical charge exchange  $p + H \leftrightarrow H + p$  with 15 eV protons at a density  $N_p = N_e = 7.7 \times 10^{14} \text{ cm}^{-3}$ . With a cross section of  $\sigma_{pH} \approx 3 \times 10^{-15} \text{ cm}^2$  (Fig. 3.24 of Ref. 13), the mean free path for entering cold hydrogen atoms to undergo charge exchange is  $(N_p \sigma_{pH})^{-1} \approx 2 \text{ cm}$ , with the resulting energetic atoms migrating through the hot plasma and serving as vuv absorbers.

Besides linewidths discussed so far, the relative line intensities also are useful in characterizing the plasma. From the measured  $n \rightarrow 1$  Lyman-series integrated relative line intensities  $I_{n1}$  for  $n=2$  to  $n=6$ , corrected for instrumental variations and opacity, the reduced populations  $N_n/g_n$  ( $\pm 50\%$ ) were determined using the standard relation  $I_{n1} \propto N_n f_n / g_n \lambda_{n1}^3$ , where  $N_n$  and  $g_n$  are population densities and statistical weights for level  $n$ , respectively, and  $\lambda_{n1}$  is the wavelength for the  $n \rightarrow 1$  transition. Similar results were obtained from the Balmer series for  $n=3$  to  $n=6$ , substituting  $I_{n2}$  and  $\lambda_{n2}$  above, with an accuracy of  $\pm 30\%$ . The measured values are plotted for both series in Fig. 3 versus  $n$  and closely follow a  $n^{-6}$  dependence. Such a dependence has been shown (in Fig. 4.5 of Ref. 13) for  $T_e=11$  eV to be characteristic of a “purely-ionizing” plasma at our measured electron density of  $7.7 \times 10^{14} \text{ cm}^{-3}$ . Here collisional ionization dominates. This is in contrast to a “purely-recombining” plasma where the opposite is true and which also is plotted in Fig. 3 for  $T_e=11$  eV, where  $N_n/g_n$  follows (see Fig. 4.13 of Ref. 13) a nearly flat distribution with  $n$ . In such a purely-ionizing plasma, ionization proceeds stepwise or “ladderlike” through excitation to higher bound states, followed then by ionization.

In summary, vacuum ultraviolet spectroscopic measurements on the Lyman resonance lines of atomic hydrogen indicate a peak opacity of  $\tau_H=20$  on the  $n=2$  to  $n=1$  line. Besides being useful in radiative transfer modeling, this has led to an average value along the line of sight for the ground-state hydrogen density  $\sim 6\%$  of that of the electrons (as well

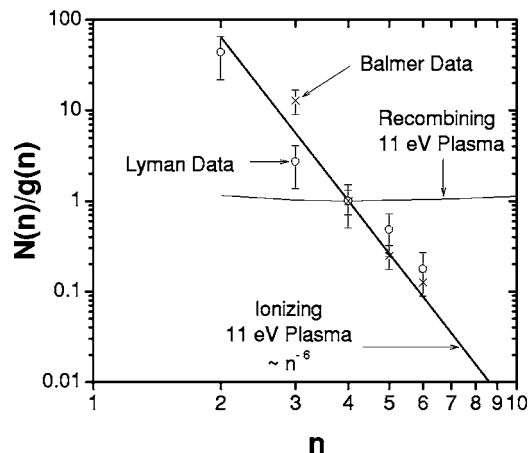


FIG. 3. Reduced population density vs upper quantum level  $n$ , showing the characteristic  $n^{-6}$  dependence of a purely ionizing plasma. Both modeling (Ref. 13) (lines) at  $T_e=11$  eV and present data (points) are plotted. Shown for comparison is an approximately constant population predicted for recombination, also at  $T_e=11$  eV. Normalization is to  $n=4$  for clarity.

as protons), determined to be  $7.7 \times 10^{14} \text{ cm}^{-3}$  from Stark broadening of visible lines and supported by interferometry. The Doppler width giving the best fit to the vuv data indicates a temperature for the hydrogen lines of  $\sim 15$  eV, in agreement with earlier measurements<sup>1</sup> for ions and electrons. The transient radiating atoms in such a hot plasma are thought to be formed by charge transfer with heated protons.

## ACKNOWLEDGMENTS

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